

Let K be a convex set. A **face** of K is a nonempty ^{convex} set $F \subset K$ such that if $x \in F$ and x is a positive combination of $x_1, \dots, x_n \in K$, then $x_1, \dots, x_n \in F$.

Equivalently, if $x \in F$ and $x = \frac{1}{2}(x_1 + x_2)$ with $x_1, x_2 \in K$, then $x_1, x_2 \in F$.

Prop: Let K be a convex set, $x \in K$.

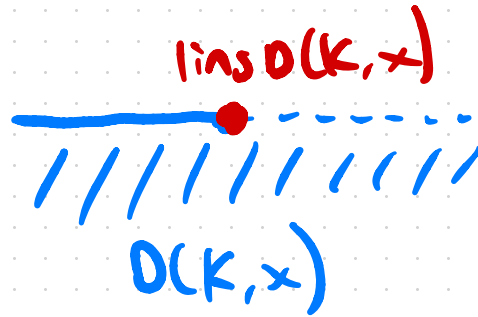
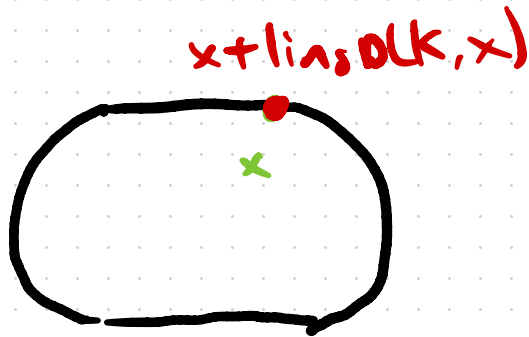
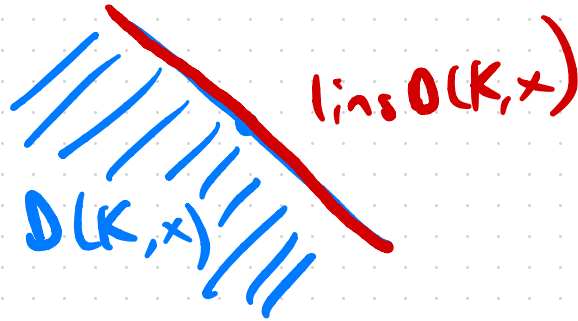
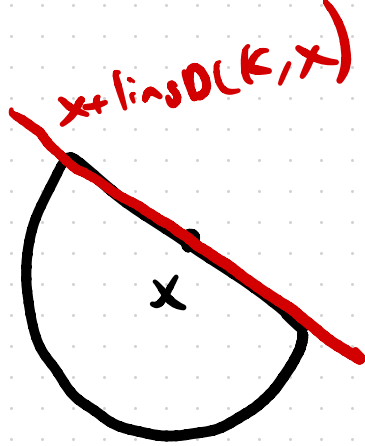
(a) There is exactly one face F of K such that $x \in \text{ri} F$.
Moreover, if G is another face containing x , then $G \supset F$.

(b) The face from part (a) is

$$F = K \cap (x + \text{lin} D(K, x))$$

$$\text{aff } F = x + \text{lin} D(K, x).$$

$$D(K, x) = \text{cone}(K - x) = \{ v \in V : x + \lambda v \in K \text{ for some } \lambda > 0 \}$$

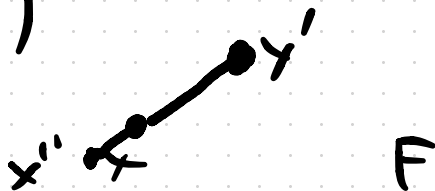


Proof:

1. There is at most one face F such that $x \in \text{int} F$, and any face G containing x must contain this F if it exists.

Proof: Suppose $x \in \text{int} F$ for some face F . Assume $x \in G$ for some other face G . Want to show $G \supset F$. Let $x' \in F$ and $x' \neq x$. Because $x \in \text{int} F$, there exists some x'' s.t. $x \in (x', x'')$

$\supset F$



Then x is a positive convex combination of x' and x'' .

Since G is a face, and $y', y'' \in K$, then $x', x'' \in G$.

In particular $x' \in G$. So $G \supset \bar{F}$, as desired.

If $x \in \text{rint } G$, then the same argument shows $F \supset G$, so $F = G$.

2. The set

$$F := K \cap (x + \text{lin } D(K, x))$$

is a face, $x \in \text{rint } F$, and $\text{aff } F = x + \text{lin } D(K, x)$.

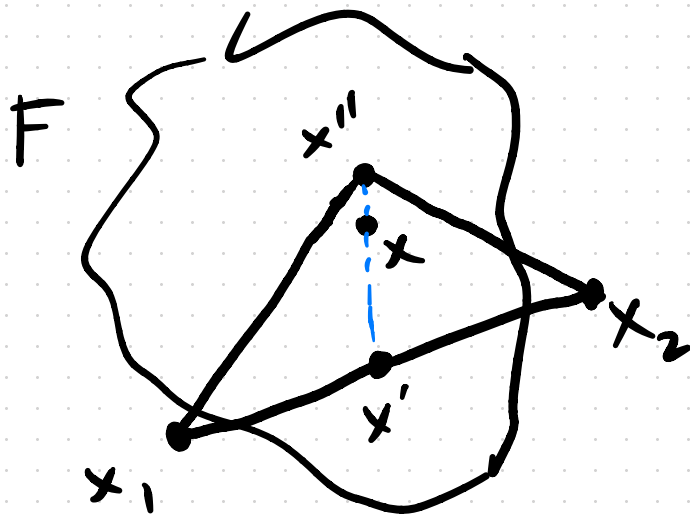
Proof: $F \subset x + \text{lin} S D(K, x)$.

$$\text{lin} S D(K, x) = \{ v \in V : x + \lambda v, x - \lambda v \in K \text{ for some } \lambda > 0 \}$$

So there is a ball of $x + \text{lin} S D(K, x)$ centered at x contained in K . By definition, this ball is also contained in $\bar{F}$. This implies that $F = x + \text{lin} S D(K, x)$ and $x \in \text{int} F$.

It remains to prove F is a face. Let $x' \in F$.

Suppose $x' = \frac{1}{2}(x_1 + x_2)$, $x_1, x_2 \in K$.



Since $x \in \text{rint } F$, there exist some $x'' \in F$ such that $x \in [x', x'']$. Let $T = \text{conv}(x_1, x_2, x'')$. By convexity $T \subset K$. Also, $x \in \text{rint } T$. Thus, there is a neighborhood of T around x . Thus, $T - x \subset \text{int } D(K, x)$.

Then $T \subset X + \text{lin} D(k, x)$, so $x_1, x_2 \in X + \text{lin} D(k, x)$.
So $x_1, x_2 \in F$. Thus F is a face.

Carollay: The relative interiors of the faces of a convex set partition the set.

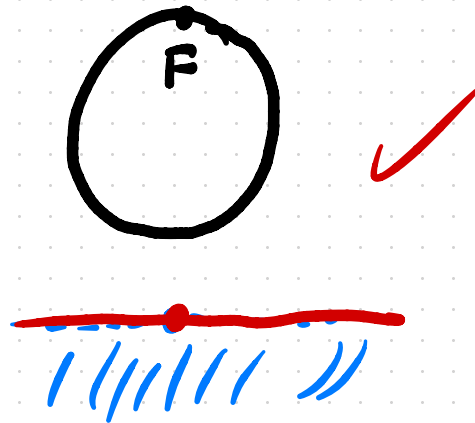
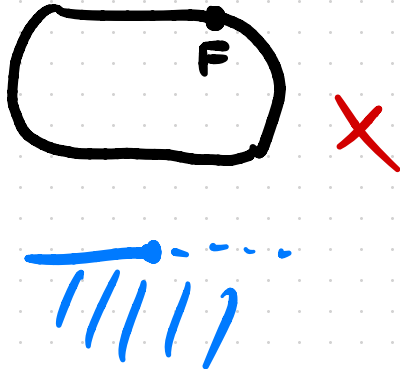
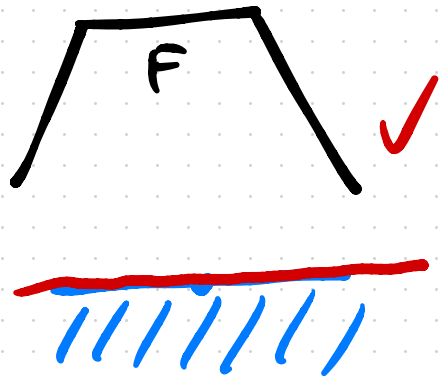
also called
"cone of feasible directions"
↓

Note: If $x, x' \in \text{ri} F$, then $D(K, x) = D(K, x')$.

Thus we can define $D(K, F) = D(K, x)$ for any $x \in \text{ri} F$

A **support set** of K is a nonempty set of the form $\operatorname{argmax}_K y$, where $y \in V^*$. (Also called **exposed faces**.)

Prop: A face F of K is a support set iff $D(K, F)$ has balanced support.

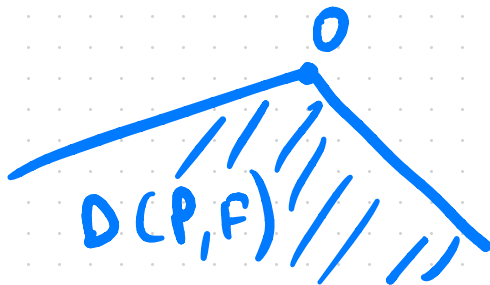
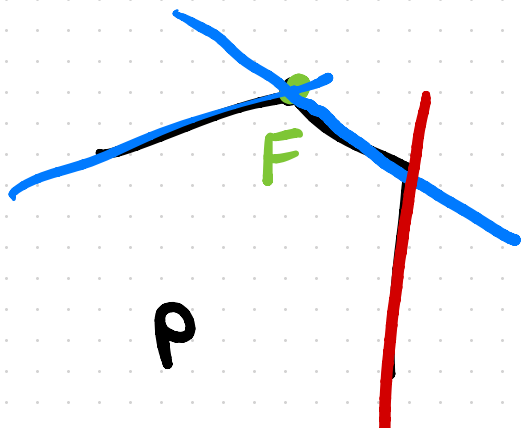


Prop: Let

$$P = \{x \in V : y_i(x) \leq b_i, i \in I\}$$

be a polyhedron, I finite. Let F be a face of P , and let $J \subset I$ be the set of all j such that $y_j(x) = b_j$ for all $x \in F$. Then

$$D(P, F) = \{v \in V : y_j(v) \leq 0, j \in J\}$$



Proof: Let $x_0 \in \text{rint } F$, so $D(P, F) = D(P, x_0)$.

First suppose $v \in D(P, F)$. Then $x_0 + \lambda v \in P$ for some $\lambda > 0$. Then for all $i \in I$, $y_i(x_0 + \lambda v) \leq b_i$

$\Rightarrow \lambda y_i(v) \leq -y_i(x_0) + b_i$. If $j \in J$, then

$y_j(x_0) - b_j = 0$, so $\lambda y_j(v) \leq 0 \Rightarrow y_j(v) \leq 0$.

so

$$D(P, F) \subset \{ y_j(x) \leq 0, j \in J \}$$

Corollary: Every face of a polyhedron is a support set.